Lecture 43 — Cumulative final review

Fri Dec 11

Last updated: October 8, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Full review day for the Final exam (Wednesday, December 16, 10:20 a.m.–12:10 p.m., FR 201). **No new quiz today** (labs/quizzes stay on their Friday schedule elsewhere). Related objectives: Lectures. Schedule.

Prepare before class

1. Final study guide — mark each outcome confident / shaky / blank.
2. Bring your cumulative definitions/formulas list and questions from quizzes, labs, and midterm practice.
3. Start (or continue) the final practice packet; we will work mixed items in class (especially Section D).

Learning objectives

By the end of class, students should be able to:

1. Connect vectors, systems, transformations, dynamics, eigenvectors, least squares, and SVD.
2. Select an appropriate method for a new situation.
3. Decide what should be done by hand and what should be delegated to software.
4. Interpret mathematical and computational results.
5. Create a targeted final-exam study plan.

In class

- Quick scan of cumulative themes (Unit I → Unit II → eigen / least squares / SVD).
- Mixed practice from the final practice packet (definitions, computations, harder mixed items).
- Partner or board work on shaky topics from the study guide checklist.
- Exit: three concrete study actions before December 16.